Data Presentation

Exam code: 9MA0

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Data presentation

What graphs and diagrams should I be familiar with?

- You will be expected to be able to use a variety of graphs such as:
  - **Box plots**
    - Can be used with ungrouped data of a single variable
    - Shows the range, interquartile range and quartiles clearly
    - Very useful for comparing data patterns quickly
  - **Cumulative frequency graphs**
    - Can be used with continuous grouped data of a single variable
    - Shows the running total of the frequencies that fall below the upper bound of each class
  - **Histograms**
    - Can be used with continuous grouped data of a single variable
    - Can be used with varying group sizes
    - Shows the frequencies of the group, represented by the area of each bar
  - **Frequency polygon**
    - Displays the shape of continuous grouped data

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

        - Can be used with ungrouped **bivariate** data (data for two variables)
        - Shows the pattern of the relationship between the variables
- You will not be expected to draw the graphs from scratch but you may be asked to add to a graph
    - You should make sure you know how to draw them anyway as it will help strengthen your understanding of the graphs

- Prior knowledge of some other 'unfamiliar' diagrams such as interpreting a stem and leaf diagram is expected
    - Review statistical diagrams from GCSE and make sure you remember how to use them
    - If there is an unfamiliar diagram in your exam, you will be able to work out how to read it by looking at the scales and labels carefully

What should I look out for when interpreting graphs?

- Look carefully at the context of the information given in the graph
- Check the scales on both axes carefully, including units
    - Sometimes the numbers will be abbreviated to fit on the scale, for example if a population is given in millions then the number 60 will represent 60 000 000
    - Look carefully at the labels and units to determine how a value should be read

- If there is more than one graph represented on the same set of axes take extra care to ensure you are reading from the correct one
- Beware of misleading graphs, the scales on the axes, units and representation can be manipulated to make a graph look more/less convincing

Worked Example

A student is collecting information on his friends' interests and believes that his friends who only have dogs spend more time outside than his friends who only have cats. He has surveyed 20 friends with only cats and 20 friends with only dogs and has written down the total amount of time, rounded to the nearest hour, each of them spent outside last week. Describe, with a reason, which diagram would be best for the student to use to display the data.

Answer:

- The data is raw and ungrouped
- Time, rounded to the nearest hour, is discrete
- The student wants to compare the distributions

The student should use two box plots to display the data, one for his friends with cats and one for his friends with dogs.

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Examiner Tips and Tricks

- Take the time needed when working with diagrams, they are usually 'easy marks' questions but it is common for students to rush them and make silly mistakes.

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